# Jian Han

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## Education

**MS** **The University of Tennessee, Knoxville**, Electrical Engineering — Knoxville, TN
Expected Dec 2027

- Focus: electric energy systems, power-system analysis, optimization, and control.
- Coursework: Power Systems Analysis, Electric Energy Systems, Optimization Methods, Power System Economics, Deep Learning for Power System Operation and Control, Statistics, Data Engineering.

**MS** **Northeastern University**, Computer Science — 2023 – 2025

- Coursework: Database Management Systems, Algorithms, Software Engineering, Networking, Operating Systems, Object-Oriented Design.

**BBA** **The University of Winnipeg**, Business and Administration – Accounting (4-year) — Winnipeg, Canada
July 2019 – Dec 2021

- Dean's Honour List – Student of Distinction (Nov 2020, Jan 2022).
- Member, Golden Key International Honour Society.
- Projects: accounting workshop with Microsoft Power BI, accounting master budget, entrepreneurship business plan.

**B.Eng.** **Hangzhou Dianzi University**, Electrical Engineering and Automation — Hangzhou, China
Sept 2006 – June 2010

- Provincial Excellent Graduate (May 2010).
- Relevant coursework: Power System Automation, Power Supply Technology, Motor Control.
- Design project: A Design for Reactive Power Compensation of a Power System (2009).

## Experience

**IpserLab, LLC**, Software Engineer Intern — Fort Worth, TX
Jan 2025 – May 2025
5 months

- Developed and debugged backend components with Java, PostgreSQL, RESTful APIs, and Git-based workflows in a cross-functional team.
- Worked with designers, engineers, and project managers while maintaining technical documentation.

**State Grid Zhejiang Hangzhou Fuyang Power Supply Co., Ltd. (State Grid Corporation of China)**, Electronic Information Technician, IT & Communications O&M

Hangzhou, China
Dec 2012 – Nov 2018
6 years

- Reliability performance evaluation (Jan – Apr 2018): analyzed 7,000+ annual fault work orders (7,065 in 2015 vs. 7,580 in 2017) to evaluate the distribution automation system. Using 500-ticket samples, average fault-repair response time fell from 57 to 36.9 minutes (~35%) and average fault-handling time from 88 to 67.6 minutes (~23%).
- Assessed economic, management, and social benefits, including projected reductions in outage energy loss and line losses; contributed to optimization recommendations for later upgrades.
- Urban distribution network planning and design (Jan – Nov 2016): assessed Fuyang's urban distribution network with GIS and fault-analysis tools to identify automation upgrade needs; contributed to infrastructure and communication-pathway plans for terminal deployment and substation automation; worked with engineers to map fault-prone areas into long-term automation plans.
- Distribution automation deployment and system integration (2015–2017): handled terminal access and network configuration for automation terminals and RTUs over fiber (EPON) and wireless channels; supported integration with the centralized control system for remote fault detection and automatic restoration; analyzed the communication network structure of switching stations and ring main units.
- Smart environmental regulation system for residential switching stations (Jan – Nov 2015): designed and implemented an IoT-based cloud platform monitoring SF6 gas, temperature, humidity, smoke, and oxygen in 10kV switching stations; built ARM-based data terminals and RESTful APIs for remote diagnostics and alerting. Enabled 24/7 remote inspection, reducing manual patrols by over 70% and preventing up to 40% of failures caused by condensation and flashover.
- PMS system data interface (2018): developed data interfaces integrating real-time data, records, and fault logs.
- Main participant in three process-improvement (QC) projects (2013–2015), including QR-code lifecycle tracking of IT equipment to reduce asset-ledger errors (2015).
- Network retrofits (2013–2014): migrated a regional network from SDH transport to dark fiber (Layer 2, STP) and upgraded access under four aggregation nodes to Layer 3, ending recurring outages; maintained company network and hardware.

**State Grid Zhejiang Hangzhou Fuyang Power Supply Co., Ltd. (State Grid Corporation of China)**, Substation Maintenance Technician

Hangzhou, China
Aug 2010 – Dec 2012
2 years 5 months

- 110kV substation construction: Xintong (Sep 2011 – Apr 2012), Chating (Apr – Jun 2012), and Dongzhou Phase II (Sep – Dec 2012). Installed 50MVA main transformers (assembly, oil filling, testing), 10kV switchgear lineups (21 panels per expansion), bus bridges and wall bushings, 110kV CTs, VTs, surge arresters, disconnect switches, post insulators and tubular bus, structures, lightning masts, and grounding; laid and terminated secondary cables from 10kV equipment to the control room.
- Testing and commissioning: tested and commissioned arc-suppression coil and capacitor bank systems; performed secondary circuit checks; completed SCADA point-to-point tests (telemetry, telesignaling, telecontrol) with district and city dispatch centers.
- 35kV projects: Jingu substation (20MVA 35/10.5kV, 7 feeders, hybrid static var compensation with harmonic filtering; installed communication and metering devices and the telemetry database on an XJ CBZ-8000 integrated automation system) and Shengcun DC system retrofit (new DC panel, battery bank reconfiguration, feeder load cut-over from both old DC buses).
- Substation project support (Sep 2010 – Jun 2011): supported four 35–110kV substation and industrial-customer projects with bid technical specifications, equipment and material order lists, contract review and technical briefings, receiving inspections, site coordination, and as-built documents and drawings for handover.
- Maintained and repaired substation equipment; performed HV/MV/LV electrical testing, transformer oil sampling, metering installation, and maintenance records.

## Projects

**LLM-for-AMS: Power-System Simulation, Transmission Analytics, and Verification** 2026 – present
CURENT LTB / The University of Tennessee, Knoxville

- Analyzed PJM 5-bus RTED load-growth scenarios with AMS and CLARABEL; identified the load levels where transmission and generator constraints became binding and forced broader redispatch (max LMP at Bus 2 rose from ~0.25 to 1.16 as Line 4 congestion intensified).
- Built a Python scenario-automation workflow that replaces manual load-variation studies with reproducible runs and structured CSV/JSON outputs for dispatch, line flows, LMPs, and solver status.
- Automated analytics for line loading, binding constraints, dispatch changes, LMP ranking, objective cost, and failed runs.
- Corrected generator dispatch reporting (Pmin, Pmax, utilization, status) so units are classified consistently as At Pmin, At Pmax, or Marginal.
- Developing verification for AI-generated grid analyses: simulator comparison, independent calculations, engineering-rule checks, and reproducibility tests.

**AI-Based PV-Curve Agent for Voltage-Stability Analysis** 2026
CURENT LTB / The University of Tennessee, Knoxville

- Developed and validated a workflow that runs PV/nose-curve studies from natural-language requests, reporting load margin, nose-point voltage, convergence, and plots.
- Analyzed how rising load moves buses toward voltage-collapse limits, relevant to large-load interconnection studies.

**Storm Prediction and Grid-Resilience Exposure Mapping** 2026
COSC 526 Data Engineering, The University of Tennessee, Knoxville (IEEE SciVis Contest 2026 data)

- Built a pipeline that streams 14 months of hourly DYAMOND GEOS climate simulation data through OpenVisus/NSDF and derives storm features (wind, pressure and temperature anomalies, vertical motion, precipitation proxy).
- Detected storm hotspots with DBSCAN; compared persistence, KNN, logistic regression, and random forest for next-day storm prediction. Logistic regression reached F1 0.712 (CSI 0.553, POD 0.863), evaluated with bootstrap CIs and McNemar's test.
- Mapped storm exposure to a grid-impact priority ranking of 36 Southeast-oriented regional patches; 24 unit tests lock the design formulas.

## Publications

**Topology-Aware Propagation-Based Assessment of Extreme-Weather Impacts on Distribution System Resilience** 2026
Co-author; contributed to technical review and paper preparation.
Junjie Yin, Xinyu Feng, **Jian Han**, Fangxing Li
(2026 North American Power Symposium (NAPS))

## Skills

**Power-System Analysis**
AC/DC power flow, DCOPF, RTED, LMP, transmission congestion, line and generator constraints, voltage stability (PV curves), sensitivity analysis, AMS, CLARABEL

**Planning & Reliability**
Load-growth scenario studies, network capacity assessment, GIS-based distribution assessment, fault-prone-area mapping, fault and outage data analysis

**Substation & Field**
35–110kV substation construction, testing and commissioning, secondary circuit checks, SCADA point-to-point testing, HV/MV/LV testing, bid specifications, as-built drawings, distribution-automation communications

**Python & Data**
Python, pandas, NumPy, SQL, PostgreSQL, MySQL, scenario automation, numerical validation, CSV/JSON processing

**Software**
Java, C, RESTful APIs, Git/GitHub, Jupyter, LangGraph, AutoCAD familiarity

## Certificates

**Electrical Engineer Professional Title** Dec 2016
State Grid Corporation of China

**AWS Academy Cloud Architecting** Nov 2024
Amazon Web Services

## Patents

- An Optical Fiber Composite Low-voltage Cable (OPLC) Cross Connecting Cabinet. China, 2014. Patent No. ZL 2014 2 0634381.4.
- An Obstacle Avoidance Remote Monitoring Data Transmission Equipment. China, 2014. Patent No. ZL 2014 2 0733874.3.

## Awards

**Dean's Honour List – Student of Distinction** Jan 2022
The University of Winnipeg

**Dean's Honour List – Student of Distinction** Nov 2020
The University of Winnipeg

**Safety Production Advanced Individual** 2013
State Grid Fuyang Power Supply Co.

**Provincial Excellent Graduate** May 2010
Hangzhou Dianzi University